MIL-E-46511A (PA)
July 25, 1975
SUPERSEDING
MIL-E-46511
30 March 1961 and
NOTICE 1
27 April 1973

MILITARY SPECIFICATION

Explosive Harness Assembly Missile Rocket, XM84 Loading Assembly and Packing

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of explosive harness assembly designated as Explosive Harness Assembly, Missile Rocket, XM-84.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions General Specifications for

STANDARDS

MILITARY

MIL-STD-453 - Inspection, Radiographic
MIL-STD-650 - Explosive: Sampling, Inspection and Testing

DRAWINGS

8850255 - Explosive Harness Assembly, Missile Rocket, XM 84
8799615 - Box, Packing, Ammunition, For Explosive Harness, Assembly, Missile Rocket, XM84

FSC: 1336

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8799615 - Box, Packing, Ammunition, For Explosive Harness, Assembly, Missile Rocket, XM84
8799616 - Carton, Packing, Ammunition, For Explosvie Harness Assembly, Missile-Rocket, XM84

2.2 Other Publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Code of Federal Regulations

Title 49 - Transportation, Parts 100-199

3. REQUIREMENTS

3.1 Materials.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and Assemblies.-The components and assemblies shall comply with all requirements specified on drawing (dwg) 8850255 and associated drawings and with all requirements specified in applicable specifications and standards.

3.2.1 Moisture Content of Pentaerythrite Tetranitrate (PETN).-The PETN moisture content shall not exceed that specified on the applicable drawing.

3.2.2 Discontinuity of Explosive Charges.-The discontinuity of charges shall not exceed that specified on the applicable drawing.

3.3 Functioning.-The explosive harness assembly shall function satisfactorily as evidenced by complete detonation of the detonating fuse assembly.

3.4 Workmanship.-

3.4.1 Workmanship-Explosive Charges.- The explosive charges shall be free of dirt, grease, chips, gouges, inhibitor solvent drippings, cracks, fissures, porosity and other defects which would make the item unsuitable for the intended use.

3.4.2 Workmanship-Inert Parts.-The parts and assemblies shall be free from chips, dirt, grease and other foreign matter that may impair proper functioning. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspections and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections.-The following types of inspections shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Relay Assembly	9298136	15
Detonating Fuse	8850262	15
Detonating Fuse Assembly	8850261	15
Forward Lead Assembly and Aft Lead Assembly	8850256	15

4.3.2 Inspections to be Performed.-See Mil-A-48078 and Table I specified herein.

4.3.3 Rejection.-See Mil-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
	Explosive Harness Assembly, Missile Rocket, XM84		8850255
			NEXT HIGHER ASSEMBLY 8850256

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Relay Assembly (Dwg. 9298136)	15	-		
	Examine for Defects			3.2	4.4.2.3
	Moisture Test			3.2.1	4.5.2
	Detonating Fuse (Dwg. 8850262)	15	-		
	Examine for Defects			3.2	4.4.2.1
	Detonating Fuse Assembly (Dwg. 8850261)	15	-		
	Examine for Defects			3.2	4.4.2.2
	Radiographic Examination for Gap			3.2.2	4.5.3
	Forward Lead Assembly and Aft Lead Assembly (Dwg. 8850256)	15	-		
	Examine for Defects			3.2	4.4.2.4
	Functioning Test			3.3	4.5.4

NOTES:

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4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot of Explosive Harness Assemblies shall contain:

a. Parts of the same interfix lot number from one manufacturer.
b. Detonating fuses from not more than one lot.
c. PETN from not more than one lot.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling Plans.-Unless otherwise specified in Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-46511A

PARAGRAPH	TITLE	SHEET 1 OF 6	DRAWING NUMBER
4.4.2.1	Detonating Fuse		8850262
			NEXT HIGHER ASSEMBLY 8850261

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131.	Squareness of ends		0.40%	3.2	Visual
Minor					
201.	Varnish missing		0.65%	3.2	Visual
202.	Evidence of poor workmanship		0.65%	3.4.1	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-46511A

PARAGRAPH	TITLE		SHEET 2 OF 6		DRAWING NUMBER 8850261
4.4.2.2	Detonating Fuse Assembly				NEXT HIGHER ASSEMBLY 8850256
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131.	Diameter of assembly, max.		0.40%	3.2	Gage
132.	Crimp improper		0.40%	3.2	Visual
133.	Relay cup damaged		0.40%	3.2	Visual
134.	Fuse cracked		0.40%	3.2	Visual
135.	Radiographic Inspection of charge		100%	3.2.2	4.5.3
Minor					
201.	Diameter of fuse		0.65%	3.2	Gage
202.	Crimp location improper		0.65%	3.2	Gage
203.	Evidence of poor workmanship		0.65%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 3 OF 6	DRAWING NUMBER
4.4.2.3	Relay Assembly		9298136
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131.	Height of charge, max.		0.40%	3.2	Gage
132.	PETN Moisture Test		100%	3.2.1	4.5.2
Minor					
201.	Diameter				
202.	Length of relay		0.65%	3.2	Gage
203.	Evidence of poor workmanship		0.65%	3.4.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 4 OF 6	DRAWING NUMBER 8850256
4.4.2.4	Forward Lead Assembly and Aft Lead Assembly		NEXT HIGHER ASSEMBLY 8850255

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	NOne defined				
Major B					
131.	Projection of relay assembly		0.40%	3.2	Gage
132.	Overall length		0.40%	3.2	Gage
133.	Diameter, max.		0.40%	3.2	Gage
134.	Crimp improper		0.40%	3.2	Visual
135.	Threads damaged		0.40%	3.2	Visual
136.	Functioning Test	(a)	1.0%	3.3	4.5.4
Minor					
201.	Tag missing or incorrect		0.65%	3.2	Visual
202.	Marking improper		0.65%	3.2	Visual
203.	Evidence of poor workmanship		0.65%	3.4.1	Visual

NOTES:

(a) Sample size per MIL-STD-105, ltr F

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 5 OF 6		DRAWING NUMBER
4.4.2.5	Packing Carton				8799616
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131.	Carton damaged		0.40%	3.2	Visual
132.	Sealing improper		0.40%	3.2	Visual
Minor					
201.	Component missing		0.65%	3.2	Visual
202.	Contents loose		0.65%	3.2	Manual
203.	Packing damp		0.65%	3.2	Visual
204.	Marking improper		0.65%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-46511A

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Packing Box for Missile Rocket, XM84	6 OF 6	8799615
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201.	Hardware missing, loose or broken		0.65%	3.2	Visual
202.	Strapping missing or loose		0.65%	3.2	Visual
203.	Box damaged		0.65%	3.2	Visual
204.	Contents loose		0.65%	3.2	Visual
205.	Car seal improper		0.65%	3.2	Visual
206.	Marking improper		0.65%	3.2	Visual

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4.4.3 Testing.-Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and Section 6 herein.

4.5 Test Method and Procedures

4.5.1 Materials, Components and Processes.-Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies and materials)have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and Mechanical Inspections.-The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Section 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 PETN Moisture Content.-The moisture determination shall be conducted in accordance with MIL-STD-650, method 101.5.

4.5.3 Discontinuity of Explosive Charges.-The Detonating Fuse assembly shall be radiographed in accordance with MIL-STD-483 to determine that gap between the explosive charge of the detonating fuse and the explosive charges of the relay assemblies. The penetrameter shall be a qualified defective detonating fuse assembly or a simulated inert unit. The penetrameter shall exhibit the maximum permissible detonating fuse-relay assembly gap. Any gap exceeding the requirements of dwg. 8850261 shall be classed defective.

4.5.4 Functioning of Explosive Harness.-A number 8 blasting cap or any high order initiating device acceptable to the procuring activity shall be fastened to the relay assembly at the connector end of the explosive harness assembly and initiated to function the detonating fuse. A detonating fuse which does not function along its entire length including the relay assembly at the nut end of the explosive harness shall be classed defective.

5. PREPARATION FOR DELIVERY

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5.1 Packaging.-Packaging shall be as specified on dwg. 8799616. Level A shall apply.

5.2 Packing and Marking.-Packing and marking shall be as specified on dwg. 8799615 and MIL-STD-1169. Level A shall apply.

6. NOTES

6.1 Intended Use.-The components covered by this specification are intended for use on the Explosive Harness Assembly, Missile Rocket, XM84 and Warhead, GM, HE, M17A1.

6.2 Ordering Data.-See MIL-A-48078.

6.3 Submission of Inspection Equipment Designs for Approval.-See MIL-A-48078. Submit equipment designs as required to Commander, Picatinny Arsenal, Attention: SARPA-QA-T, Dover, N.J. 07801.

Custodian:
Army-PA

Preparing Activity
Army-PA

Project Number: 1136-A253

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